

FORM 2
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COMPLETE SPECIFICATION
(See section 10; rule 13)

Title: Systems and Methods for Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction

APPLICANT DETAILS:

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PREAMBLE TO THE DESCRIPTION:

The following specification (particularly) describes the nature of the invention (and the manner in which it is to be performed):

Systems and Methods for Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction

Field of the Invention

The present invention relates to apparatus, Systems and Methods for Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction and Diagnosis Networks.

Background of the Invention

The following background discussion includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

Wearable or portable health monitoring devices are designed to continuously collect, process, analyse, store and transmit the collected data and extracted clinical parameters. Due to the simple and ease of sensing, the PPG sensor(s) is/are embedded with health monitoring devices and fitness monitors to extract vital signs and also detect life threatening arrhythmias. Most health monitoring devices and vital signs monitors produce often false alarms under different kinds of noisy PPG signals and also recording scenarios, that can lead to wrong cardiac diagnosis and also lead to more energy consumption which can reduce lifetime of battery in addition to increasing the transmission and treatment costs when normal PPG signals are transmitted continuously to remote computing platforms for further analysis at the remote or edge computing server. For example, WO2017089921A1 discloses a system and method that facilitate automatically identifying and selecting clean segments of the PPG signal before PPG-derived parameters (e.g., pulse transit time, heart rate) are calculated and used in clinical decision support algorithms, thereby overcoming the above-referenced problems and others.

The gap in existing health prediction and data analysis methods (prior arts) and commercial health monitoring products are: more false alarms, poor diagnostic accuracy, demand-aware health computing and climate information fused health prediction.

Health monitoring devices often lead to false alarms and irrelevant diagnostic due to the absence of contextual information in the case of semi-supervised and unsupervised health

monitoring applications and also remote health monitoring networks including, telehealth, mobile-health, telemedicine, home health monitoring, and also wearables.

Hence, there is needed an effective system and method for contextual information aware human disease diagnosis and prediction that can enable accurate and reliable diagnosis and prediction of diseases with reduced false alarms by automatically fusing and finding clinical-relevance of contextual information that are new innovative ideas to address major limitations of the existing health monitoring devices or products.

Object(s) of the Invention:

A primary object of the present invention is to overcome the drawbacks associated with the prior art.

Yet another object of the present invention is to provide a system and method for contextual information aware human disease diagnosis and prediction that can enable accurate and reliable diagnosis and prediction of diseases with reduced false alarms by automatically fusing and finding clinical-relevance of contextual information that are new innovative ideas to address major limitations of the existing health monitoring devices or products.

Summary of the invention:

In an aspect of the present invention there is provided a system for contextual information aware physiological parameter analysis and intelligent human disease prediction comprising the components of:

- Embedded sensors for simultaneously sensing physiological signals and non-cardiac signals comprising contextual signals;
- Distributed sensors for extracting contextual information from the internet sources and extracting other parameters from the acquired signals at different levels of computing including the on-device, edge, mist, fog and cloud computing wherein the selection of computing platforms comprising on-device, edge, mist/fog and cloud, is based on the health computing requirement for predicting health risk factors and also computational resources availability;
- Configurable physiological signal sensing module,

- Contextual information sensing module,
- Quality-aware physiological signal processing module,
- Physical activity sensing module,
- Weather information collection module, by collecting the data from online weather information portal and the embedded sensors, wherein the system collects the weather information comprising different surrounding pollutants, extreme heat and extreme cold, humidity and atmospheric pressure related to the physiological changes which can vary vital signs parameters.
- Physiological parameter and contextual information fusion module,
- human attributes collection module,
- Disease prediction layer and health risk recommendation module
- Deep learning context-aware disease prediction network module.

wherein the system comprises configurable physiological signal sensing, weather and emotional sensing in plug-sense mode and also variable sampling rate system that can be adopted based on type of physiological parameters need to be measured and also the type of disease to be monitored.

In another aspect of the present invention there is provided a method of operation of the system as described above comprising the steps of:

- a) human activity recognition based on the movement and acceleration signals with changes of location parameters wherein the device performs following functions in the case of low-resourced health and fitness monitoring devices:
 - i) extraction of features from the signals and combination with data for the plurality of parameters like human activities including the sitting, standing, lying (sleeping), walking, jogging, running, skating, jumping, skipping, climbing up, and clamping down;
 - ii) applies the deep learning based activity recognition method with raw and spectral images for the parameters obtained in step (i) for recognising the human activities.
- b) the device performs combination of deep learning networks with the optimal parameters for predicting diseases and health risk factors, in case of higher computational resourced platform.

This together with the other aspects of the present invention along with the various features of novelty that characterized the present disclosure is pointed out with particularity in claims annexed hereto and forms a part of the present invention. For better understanding of the present disclosure, its operating advantages, and the specified objective attained by its uses, reference should be made to the accompanying descriptive matter in which there are illustrated exemplary embodiments of the present invention.

Detail Description of the Drawings

To further clarify advantages and features of the present invention, a more particular description of the invention will be rendered by reference to specific embodiments thereof, which is illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope. The invention will be described and explained with additional specificity and detail with the accompanying drawings in which:

Figure 1: Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction and Diagnosis Apparatus

Figure 2: Human Activity Recognition Methods for Lower and Higher Computational Resource Platforms

Figure 3: Quality-Aware Physiological Parameter Extraction Methods for Lower and Higher Computational Resource Platforms

Figure 4: Sound Activity Recognition Methods for Lower and Higher Computational Resource Platforms

Figure 5: Music Recognition Methods for Lower and Higher Computational Resource Platforms

Figure 6: Weather Information Collection Method Using Online Weather Information Portal or Embedded Sensors

Detailed Description of the Invention

For the purpose of promoting an understanding of the principles of the invention, reference will now be made to the embodiment illustrated in the drawings and specific language will be used

to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended, such alterations and further modifications in the illustrated system, and such further applications of the principles of the invention as illustrated therein being contemplated as would normally occur to one skilled in the art to which the invention relates.

It will be understood by those skilled in the art that the foregoing general description and the following detailed description are exemplary and explanatory of the invention and are not intended to be restrictive thereof. Throughout the patent specification, a convention employed is that in the appended drawings, like numerals denote like components.

Reference throughout this specification to “an embodiment, “another embodiment, “an implementation, “another implementation” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrase “in an embodiment, “in another embodiment, “in one implementation, “in another implementation, and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

The terms "comprises", "comprising", or any other variations thereof, are intended to cover a non-exclusive inclusion, such that a process or method that comprises a list of steps does not include only those steps but may include other steps not expressly listed or inherent to such process or method.

The present invention disclosure particularly relates to health monitoring devices or vital sign monitoring devices for effective analysis and diagnosis of human diseases by combining contextual information including the physical activity, weather conditions (extreme cold and heat, humidity, pressure), environmental pollutants (like air pollutants, sound pollutants etc.), more particularly with different kinds of parameters extracted from physiological signals.

The present technology relates to wearable or portable health monitoring devices with higher false alarm reduction rate and particularly to quality-aware physiological signal processing, demand-aware computing platforms (edge, mist, fog and cloud computing), smart contextual-information gathering techniques, and artificial intelligence networks, more particularly relates to methods and systems for enabling energy-efficient signal computing (on-device, edge, mist, fog and cloud) and for improving diagnostic accuracy under supervised, semi-supervised and

unsupervised analysis and diagnosis of various human diseases, prediction of diseases, and monitoring of health risk factors.

In an embodiment, the system comprises following components/ units:

- Contextual information aware human disease prediction unit
- Deep learning based physical activity recognition network with geographical location change information unit
- Physiological and contextual information embedding unit
- Deep learning context-aware disease prediction network unit
- Demand-aware computing (edge, mist, fog and cloud computing) unit

The system performs following functions like:

- Quality-aware physiological signal processing
- Weather information collection
- Combination of Physiological parameter and contextual information
- Human attributes collection
- Disease prediction and health risk recommendation

In an embodiment, the system performs functions comprising but not limited to quality-aware physiological signal processing methods, weather information collection method, physiological parameter and contextual information fusion method, human attributes collection method, disease prediction layer and health risk recommendation method, deep learning context-aware disease prediction network system, deep learning based physical activity recognition network with geographical location change information system, physiological and contextual information embedding system, demand-aware computing (edge, mist, fog and cloud computing) system.

In an embodiment, the invention discloses a health monitoring device or vital sign monitoring device for effective analysis and diagnosis of human diseases by combining the contextual information including the physical activity, weather conditions (extreme cold and heat, humidity, pressure), environmental pollutants (like air pollutants, sound pollutants etc.), more particularly with different kinds of parameters extracted from physiological signals.

Aspects of this present invention comprises contextual information aware human disease diagnosis and prediction apparatus, and also systems and methods for simultaneously sensing

physiological signals using embedded sensors and non-cardiac signals (contextual signals) using embedded sensors or/and distributed sensors (or contextual information extracted from the internet sources and extracting various parameters from the acquired signals at different levels of computing including the on-device, edge, mist, fog and cloud computing. The proposed apparatus includes different layers, including the configurable physiological signal sensing layer, contextual information sensing layer, quality-aware physiological signal processing layer, physical activity sensing layer, weather information collection layer, physiological parameter and contextual information fusion layer, human attributes collection layer, disease prediction layer and health risk recommendation layer and deep learning context-aware disease prediction network.

The present invention discloses human activity recognition based on the movement and acceleration signals with changes of location parameters. For low-resourced health and fitness monitoring devices, the present invention uses the method of features extracted from these signals and combined with sets of decision rules to recognize various kinds of human activities including the sitting, standing, lying (sleeping), walking, jogging, running, skating, jumping, skipping, climbing up, and clamping down. For low-resourced health and fitness monitoring devices, the present invention uses the method of deep learning based activity recognition method with raw and spectral images of the activity and movement related for recognition human activities.

Since the human activity impacts variations in vital signs that are not due to the defects of anatomic structures or abnormalities of organs, the present apparatus combines the human activity information to improve diagnostic relevance and accuracy. The pulse rate or heart rate may vary with different human activities, for example, increases when the normal subject is running and decreases when the normal subject is sleeping. For abnormal subjects, the heart rate may decrease or increase due to the malfunctioning of heart organs when the subject is in resting condition (or no activities). Therefore, based on our research findings, the contextual aware disease prediction system is disclosed that can improve diagnostic relevancy. The present invention discloses time-domain features of activity signals and location change features are used to determine the non-movement or movement activity order to reduce processing time and energy consumption of high-level features based human activity recognition. If the system detects movement activity, then the present invention apparatus initiates the recognition of human activities related to movement or motion such as walking,

jogging, skating, running, climbing up and so on. If the system detects non-movement activity, then the present invention apparatus initiates the recognition of human activities related to non-movement or no motion such as sitting, standing, laying (sleeping), jumping, skipping and so on.

Certain aspects of the present invention are related to the weather information collection such as different surrounding pollutants, extreme heat and extreme cold, humidity and atmospheric pressure that related to the physiological changes which can vary vital signs parameters. The present invention collects from online weather information portal or embedded sensors. The fusion of these weather information can improve the diagnostic relevance and accuracy.

In another embodiment, the invention discloses emotional recognition based on the processed sound processing method which collects background sounds using the microphone(s) and uses dimensional and two-dimensional representation of sounds with deep learning networks for identifying type of background sounds and recognition musical classes. The present invention collects musical information from the hand-held music playing devices and also voice communication for extracting emotional related information which can be embedded or fused with physiological signals for analysis of pathological data and also predicting possible variations in functioning of the organs.

The present invention includes the signal quality assessment modules for accurately estimating vital signs and clinical indexes from the acquired physiological signals and disease prediction module by combining the extracted physiological parameters and clinical indexes and the weather information and emotional information for improving diagnostic relevance. The present invention of disease prediction is performed based on the computational resources availability. For computational resourced platform, method of disease prediction is based on the decision rules combined with extracted pathological and non-pathological parameters and also weather information. For higher computational resourced platform, deep learning networks with the proposed optimal parameters are used for predicting diseases and health risk factors.

The present invention of context-aware information based disease prediction includes configurable physiological signal sensing, weather and emotional sensing in plug-sense mode and also variable sampling rate system that can be adopted based on type of physiological parameters need to be measured and also type disease needs to be monitored.

The present invention discloses selection of computing platforms (on-device, edge, mist/fog and cloud) based on the health computing requirement for predicting health risk factors and also computational resources availability.

In an embodiment, the system for contextual information aware physiological parameter analysis and intelligent human disease prediction comprising the components of:

- a) Embedded sensors for simultaneously sensing physiological signals and non-cardiac signals comprising contextual signals;
- b) Distributed sensors for extracting contextual information from the internet sources and extracting other parameters from the acquired signals at different levels of computing including the on-device, edge, mist, fog and cloud computing wherein the selection of computing platforms comprising on-device, edge, mist/fog and cloud, is based on the health computing requirement for predicting health risk factors and also computational resources availability;
- c) Configurable physiological signal sensing module,
- d) Contextual information sensing module,
- e) Quality-aware physiological signal processing module,
- f) Physical activity sensing module,
- g) Weather information collection module, by collecting the data from online weather information portal and the embedded sensors, wherein the system collects the weather information comprising different surrounding pollutants, extreme heat and extreme cold, humidity and atmospheric pressure related to the physiological changes which can vary vital signs parameters.
- h) Physiological parameter and contextual information fusion module,
- i) human attributes collection module,
- j) Disease prediction layer and health risk recommendation module
- k) Deep learning context-aware disease prediction network module.

wherein the system comprises configurable physiological signal sensing, weather and emotional sensing in plug-sense mode and also variable sampling rate system that can be adopted based on type of physiological parameters need to be measured and also the type of disease to be monitored.

In an embodiment, the system measures time-domain features of activity signals and location change features for determining the non-movement or movement activity order to reduce processing time and energy consumption of high-level features based human activity recognition.

In an embodiment, the system detects movement activity, by recognising the human activities related to movement or motion comprising walking, jogging, skating, running, climbing up and so on.

In an embodiment, the system detects non-movement activity, by recognising the human activities related to non-movement or no motion such as sitting, standing, laying (sleeping), jumping, skipping.

In an embodiment, the system detects emotional recognition based on the processed sound processing method which collects background sounds using the microphone(s) and uses dimensional and two-dimensional representation of sounds with deep learning networks for identifying type of background sounds and recognition musical classes.

In an embodiment, the system collects musical information from the hand-held music playing devices and also voice communication for extracting emotional related information which can be embedded or fused with physiological signals for analysis of pathological data and also predicting possible variations in functioning of the organs.

In an embodiment, the signal quality assessment module accurately estimates vital signs and clinical indexes from the acquired physiological signals and disease prediction module by combining the extracted physiological parameters and clinical indexes and the weather information and emotional information for improving diagnostic relevance, followed by predicting the disease based on the computational resources wherein the computational resource platform predicts the disease based on the decision rules combined with extracted pathological and non-pathological parameters along with the weather information.

In an embodiment, the method of operation of the system as described above comprises the steps of:

- a) human activity recognition based on the movement and acceleration signals with changes of location parameters wherein the device performs following functions in the case of low-resourced health and fitness monitoring devices:
 - i) extraction of features from the signals and combination with data for the plurality of parameters like human activities including the sitting, standing, lying (sleeping), walking, jogging, running, skating, jumping, skipping, climbing up, and clamping down;
 - ii) applies the deep learning based activity recognition method with raw and spectral images for the parameters obtained in step (i) for recognising the human activities.
- b) the device performs combination of deep learning networks with the optimal parameters for predicting diseases and health risk factors, in case of higher computational resourced platform.

In an embodiment, the emotional recognition is based on the processed sound processing method which collects background sounds by the microphone(s) and uses dimensional and two-dimensional representation of sounds with deep learning networks for identifying type of background sounds and recognition musical classes.

In an embodiment, the device collects musical information from the hand-held music playing devices and also voice communication for extracting emotional related information which can be embedded or fused with physiological signals for analysis of pathological data and also predicting possible variations in functioning of the organs.

In an embodiment, Figure 1 shows the integration of activity signal acquisition, physiological signal acquisition, weather data acquisition, pollutant data acquisition and sound data acquisition. The data is collected and the contextual information aware physiological parameter analysis is performed. This analysis also includes music emotional recognition and physiological parameters.

In an embodiment, Figure 2 shows movement and Non-Movement Activity Detection for deep learning network based activity recognition and for determining the activity mapping rules.

In an embodiment, Figure 3 shows physiological parameter extraction for Low-Computational Resource Platform and high-Computational Resource Platform.

In an embodiment, Figure 4 shows sound recognition method for Low-Computational Resource Platform and high-Computational Resource Platform.

In an embodiment, Figure 5 shows music recognition method for Low-Computational Resource Platform and high-Computational Resource Platform.

In an embodiment, Figure 6 shows weather Information Collection Method Using Online Weather Information Portal or Embedded Sensors

Commercial aspects of the invention and proposed technologies are summarized below:

- AI Health Prediction Networks
- Personalized Health Prediction and Disease Diagnosis Systems
- Smart Portable or Wearable Health Monitoring Devices
- Smart Fitness Monitoring Devices
- Internet of Medical Things
- AI Healthcare Kiosks

The Invention is further described with the help of non-limiting examples:

Example 1:

Based on the evaluation results of each of the proposed methods and systems which are integral part of the Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction and Diagnosis Apparatus, the human activity recognition method had accuracy ranging from 85% to 94% for recognition movement and non-movement activities by using the waveform features and decision rules and had accuracy ranging from 89% to 98% for recognition movement and non-movement activities by using deep learning networks at the cost of higher computational resources. The signal quality assessment methods had a sensitivity of 96% to 100% for detecting noise-free physiological signal and had a specificity of 86% to 100% for detecting noises and artefacts by using waveform features and decision rules. The deep learning based

physiological signal quality assessment method with optimal convolutional neural network parameters had a sensitivity of 96% to 100% for detecting noise-free physiological signal and had a specificity of 86% to 100% for detecting noises and artefacts by using one-dimensional and two-dimensional representation of physiological signals. Based on the time-domain features, the sound activity detection method had an accuracy of 88% to 92% for different sounds of transportation and natural. The deep learning networks with optimal convolution network parameters based method had an accuracy of 88% to 92%. The music sound classification method had accuracy of 86% to 92% for five music genres using the decision rules and had accuracy of 92% to 96% for five music genres using deep learning networks at the cost of higher computational resources.

We claim:

1. A system for contextual information aware physiological parameter analysis and intelligent human disease prediction comprising the components of:
 - a) Embedded sensors for simultaneously sensing physiological signals and non-cardiac signals comprising contextual signals;
 - b) Distributed sensors for extracting contextual information from the internet sources and extracting other parameters from the acquired signals at different levels of computing including the on-device, edge, mist, fog and cloud computing wherein the selection of computing platforms comprising on-device, edge, mist/fog and cloud, is based on the health computing requirement for predicting health risk factors and also computational resources availability;
 - c) Configurable physiological signal sensing module,
 - d) Contextual information sensing module,
 - e) Quality-aware physiological signal processing module,
 - f) Physical activity sensing module,
 - g) Weather information collection module, by collecting the data from online weather information portal and the embedded sensors, wherein the system collects the weather information comprising different surrounding pollutants, extreme heat and extreme cold, humidity and atmospheric pressure related to the physiological changes which can vary vital signs parameters.
 - h) Physiological parameter and contextual information fusion module,
 - i) human attributes collection module,
 - j) Disease prediction layer and health risk recommendation module
 - k) Deep learning context-aware disease prediction network module.

wherein the system comprises configurable physiological signal sensing, weather and emotional sensing in plug-sense mode and also variable sampling rate system that can be adopted based on type of physiological parameters need to be measured and also the type of disease to be monitored.

2. The system as claimed in claim 1, wherein the system measures time-domain features of activity signals and location change features for determining the non-movement or

movement activity order to reduce processing time and energy consumption of high-level features based human activity recognition.

3. The system as claimed in claim 1, wherein the system detects movement activity, by recognising the human activities related to movement or motion comprising walking, jogging, skating, running, climbing up and so on.
4. The system as claimed in claim 1, wherein the system detects non-movement activity, by recognising the human activities related to non-movement or no motion such as sitting, standing, laying (sleeping), jumping, skipping.
5. The system as claimed in claim 1, wherein the system detects emotional recognition based on the processed sound processing method which collects background sounds using the microphone(s) and uses dimensional and two-dimensional representation of sounds with deep learning networks for identifying type of background sounds and recognition musical classes.
6. The system as claimed in claim 1, wherein the system collects musical information from the hand-held music playing devices and also voice communication for extracting emotional related information which can be embedded or fused with physiological signals for analysis of pathological data and also predicting possible variations in functioning of the organs.
7. The system as claimed in claim 1, wherein the signal quality assessment module accurately estimates vital signs and clinical indexes from the acquired physiological signals and disease prediction module by combining the extracted physiological parameters and clinical indexes and the weather information and emotional information for improving diagnostic relevance, followed by predicting the disease based on the computational resources wherein the computational resource platform predicts the disease based on the decision rules combined with extracted pathological and non-pathological parameters along with the weather information.
8. A method of operation of the system as claimed in claim 1 comprising the steps of:
 - a) human activity recognition based on the movement and acceleration signals with changes of location parameters wherein the device performs following functions in the case of low-resourced health and fitness monitoring devices:
 - i) extraction of features from the signals and combination with data for the plurality of parameters like human activities including the sitting, standing,

- lying (sleeping), walking, jogging, running, skating, jumping, skipping, climbing up, and clamping down;
- ii) applies the deep learning based activity recognition method with raw and spectral images for the parameters obtained in step (i) for recognising the human activities.
- b) the device performs combination of deep learning networks with the optimal parameters for predicting diseases and health risk factors, in case of higher computational resourced platform.
9. The method as claimed in claim 5, wherein the emotional recognition is based on the processed sound processing method which collects background sounds by the microphone(s) and uses dimensional and two-dimensional representation of sounds with deep learning networks for identifying type of background sounds and recognition musical classes.
10. The method as claimed in claim 9, wherein the device collects musical information from the hand-held music playing devices and also voice communication for extracting emotional related information which can be embedded or fused with physiological signals for analysis of pathological data and also predicting possible variations in functioning of the organs.

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ABSTRACT

Systems and Methods for Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction

The present invention relates to apparatus, Systems and Methods for Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction and Diagnosis Networks.

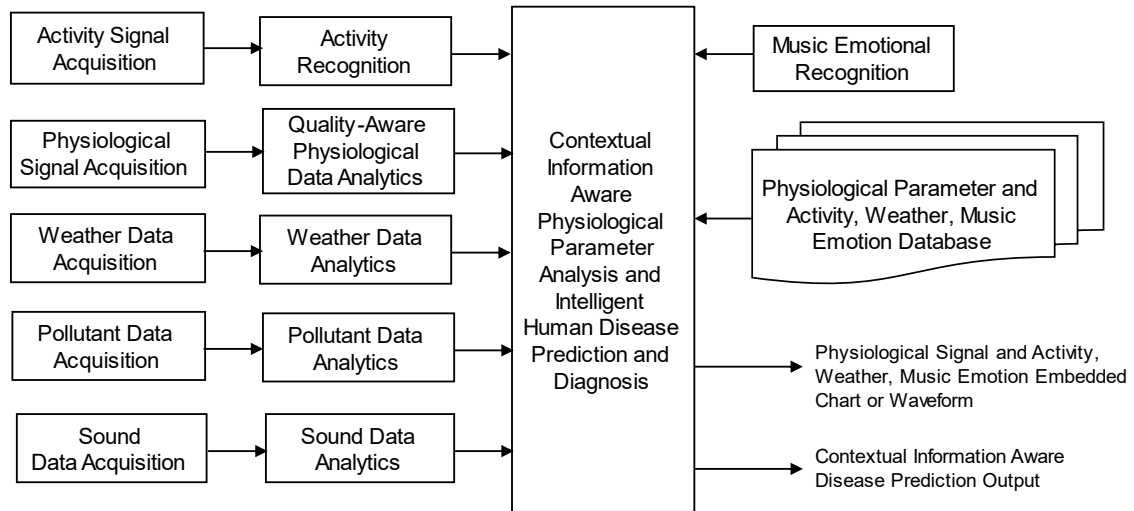
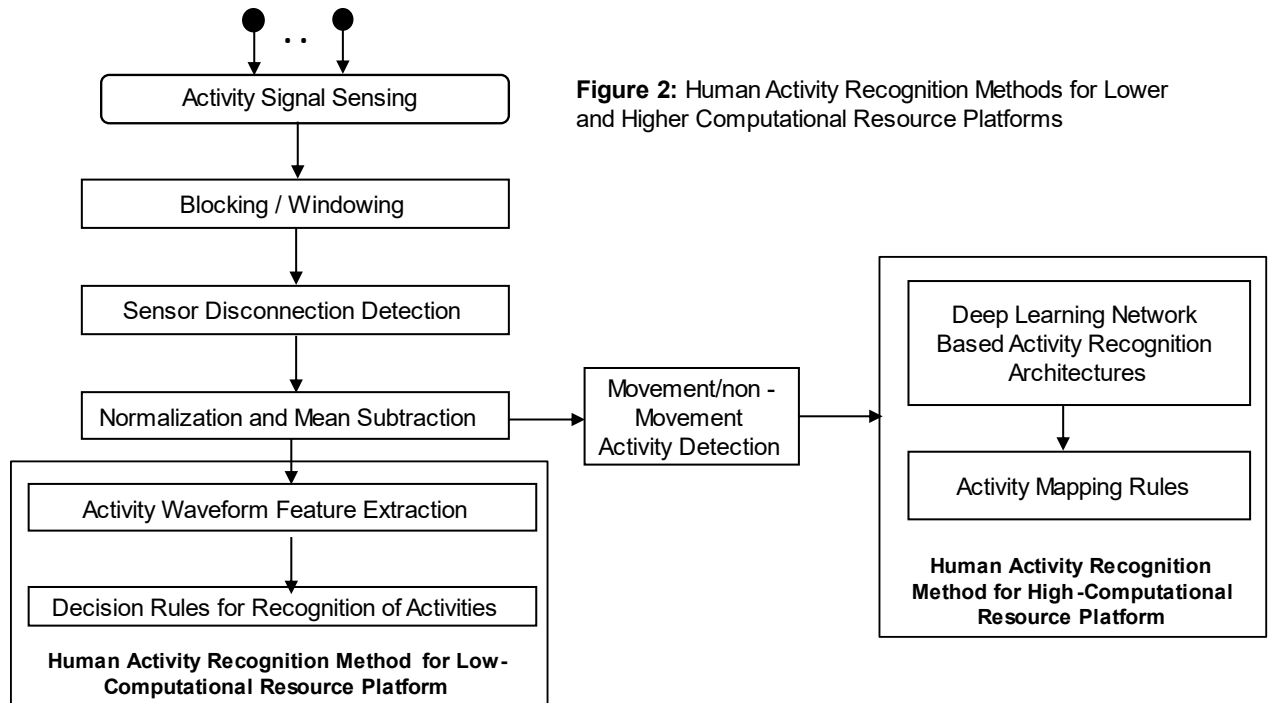
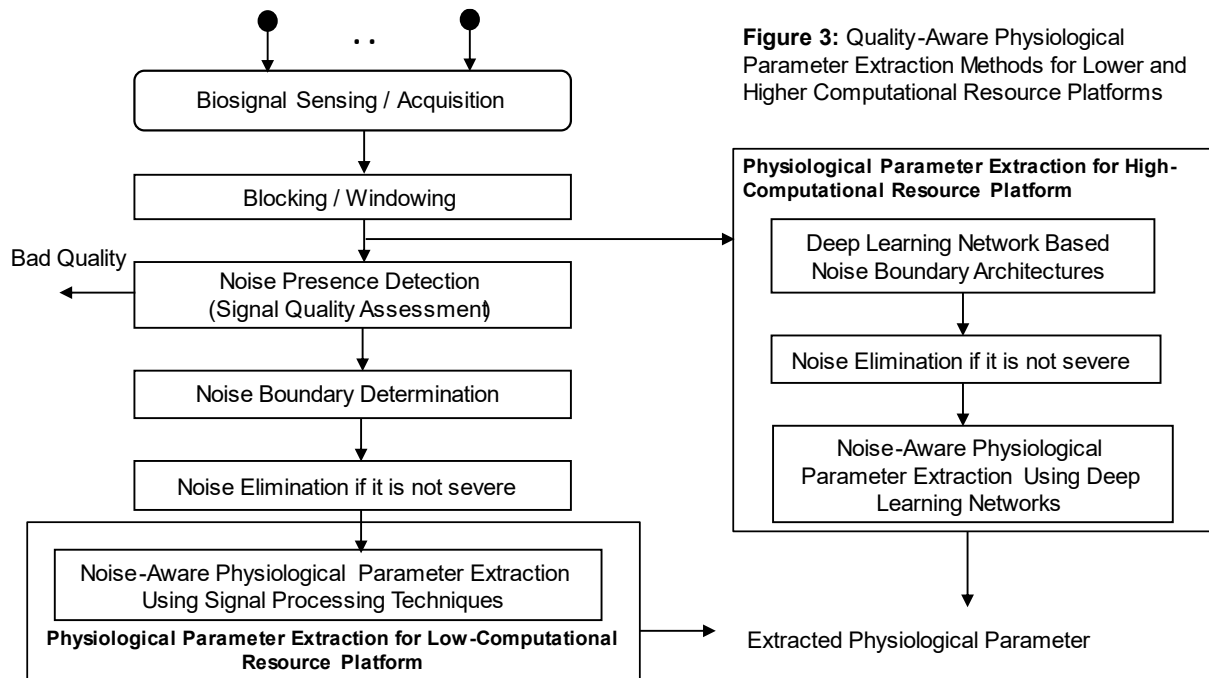


Figure 1: Contextual Information Aware Physiological Parameter Analysis and Intelligent Human Disease Prediction and Diagnosis Apparatus

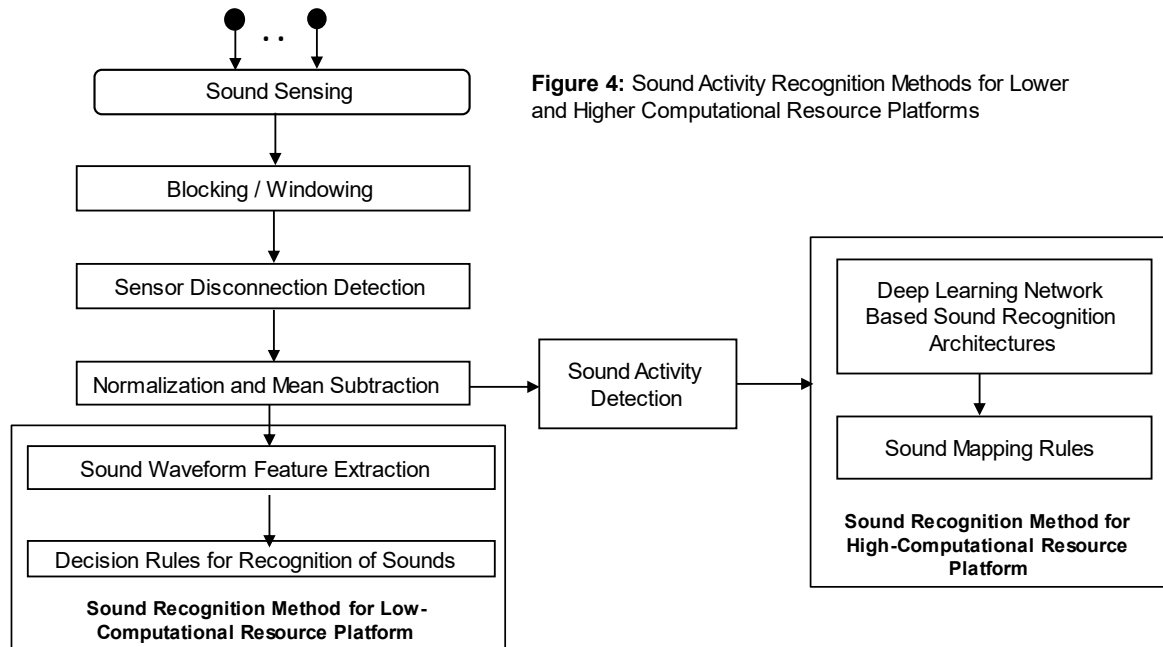
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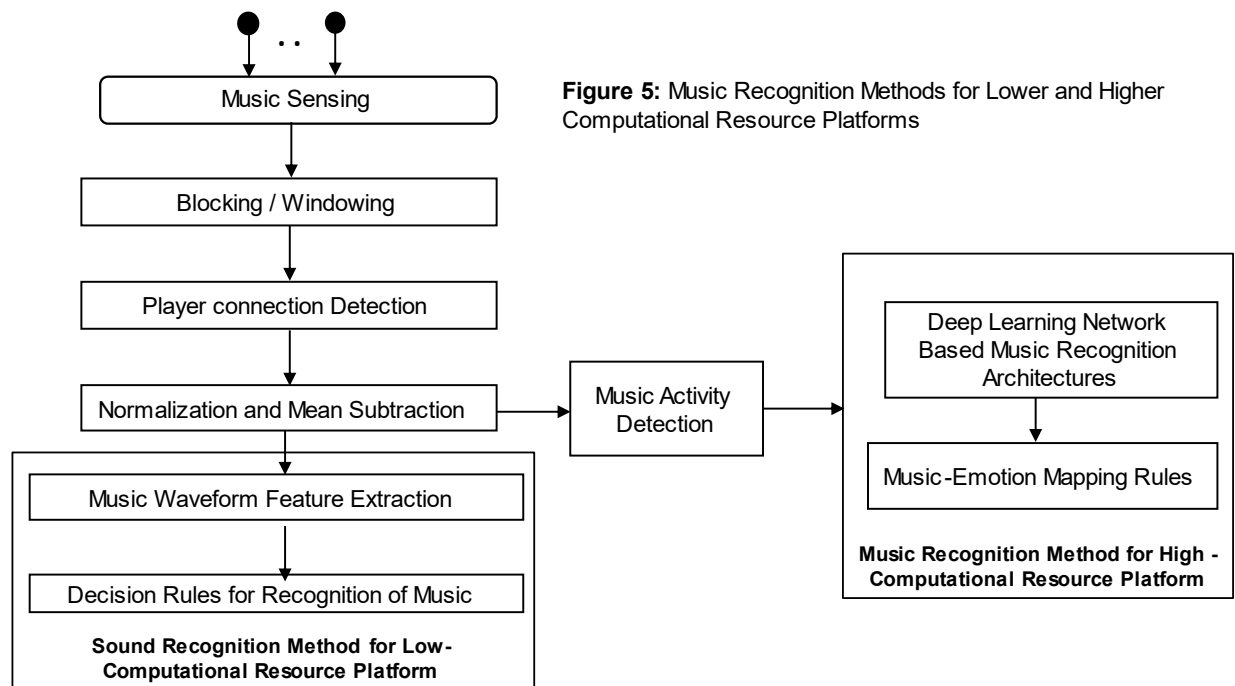
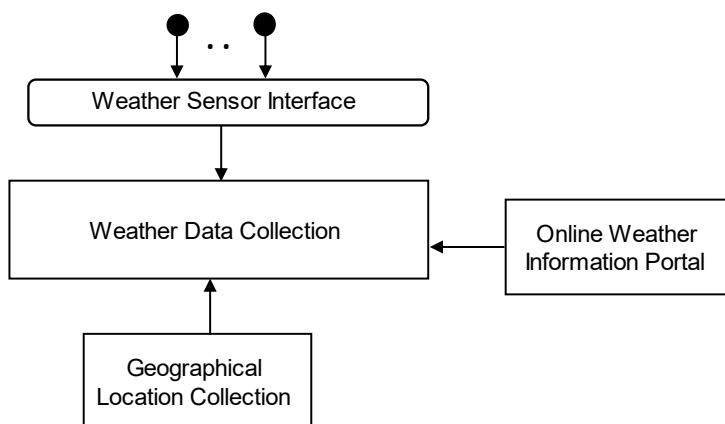


Figure 5: Music Recognition Methods for Lower and Higher Computational Resource Platforms

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Figure 6: Weather Information Collection Method Using Online Weather Information Portal or Embedded Sensors



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